



RIVER VALLEY HIGH SCHOOL

H2 CHEMISTRY

Suggested Solutions for 2019 A Level H2 Chemistry Paper

Paper 1

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	16	A
2	C	17	B
3	B	18	A
4	C	19	A
5	C	20	D
6	A	21	D
7	D	22	C
8	B	23	A
9	D	24	C
10	D	25	B
11	A	26	B
12	B	27	C
13	D	28	C
14	D	29	B
15	D	30	C

Paper 2

- 1 (a) (i) Ca: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
 Ca^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6$ [1]
- (ii) Ca^{2+} is smaller than Ca. Ca has one more filled principal quantum shell than Ca^{2+} . [1]
- (iii) Down Group 2, the effect of the increase in nuclear charge is cancelled out by the increase in shielding effect. Number of filled principal quantum shells increases. Hence, ionic radii increases down the group. [2]
- (iv) Sr has a significantly higher atomic mass compared to Ca, while the volume of Sr atom is not significantly larger than that of Ca. [1]
- (b) (i) $CuCO_3(s) \rightarrow CuO(s) + CO_2(g)$ [1]
- (ii) Ionic radius of $Cu^{2+} = 0.073 \text{ nm}$
 Ionic radius of $Mg^{2+} = 0.065 \text{ nm}$
 Ionic radius of $Ca^{2+} = 0.099 \text{ nm}$
 Minimum decomposition temperature of $CuCO_3 = 500 \text{ }^\circ\text{C}$
 (any value between $500 \text{ }^\circ\text{C}$ to $600 \text{ }^\circ\text{C}$)
Ionic radius of Cu^{2+} is between that of Mg^{2+} and Ca^{2+} , while all three ions have $2+$ charge. The smaller the cation, the higher its charge density. Hence, the electron cloud of CO_3^{2-} will be distorted to a larger extent, and C—O bonds will be weakened to a larger extent. Thus, less energy is needed to decompose $CuCO_3$ compared to $CaCO_3$. [4]
- (c) (i) 1) Compounds with copper(II) ions have catalytic properties while compounds with magnesium ions do not.
 2) Compounds with copper(II) ions are coloured while compounds with magnesium ions are not. [2]
- (ii)
 - Both can conduct electricity in the aqueous or molten state.
 - Both of their hydroxides are insoluble in excess $NaOH(aq)$.
 - Both of their oxides are basic.
 [2]

[Total: 14]

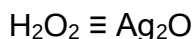
- 2 (a) (i) Hydrogen peroxide is behaving as a reducing agent. Hydrogen peroxide causes the oxidation state of silver to decrease from +1 in Ag_2O to 0 in Ag. [1]

(ii) Any one of the following:

- The volume of O_2 produced at regular time intervals could be measured to monitor the rate of reaction. The O_2 can be collected in a gas syringe/ over water in an inverted burette.
- The mass of the reaction mixture at regular time intervals could be measured by placing the setup on a weighing balance and allowing O_2 to escape into the atmosphere.
- A fixed volume of reaction mixture can be pipetted at regular time intervals and titrated with acidified KMnO_4 of a known concentration to obtain the titre (sampling method). [2]

(b) (i) amount of O_2 in $20.0 \text{ cm}^3 = \frac{20.0}{24000}$
 $= 8.333 \times 10^{-4} \text{ mol}$
 $2\text{H}_2\text{O}_2 \equiv \text{O}_2$
 amount of H_2O_2 in $1.0 \text{ cm}^3 = 2 \times 8.333 \times 10^{-4}$
 $= 0.001667 \text{ mol}$
 concentration of $\text{H}_2\text{O}_2 = 0.001667 \div \frac{1.0}{1000}$
 $= 1.667$
 $= 1.67 \text{ mol dm}^{-3}$ (to 3 s.f.) [1]

(ii) amount of H_2O_2 in $10.0 \text{ cm}^3 = 1.667 \times \frac{10.0}{1000}$
 $= 0.01667 \text{ mol}$



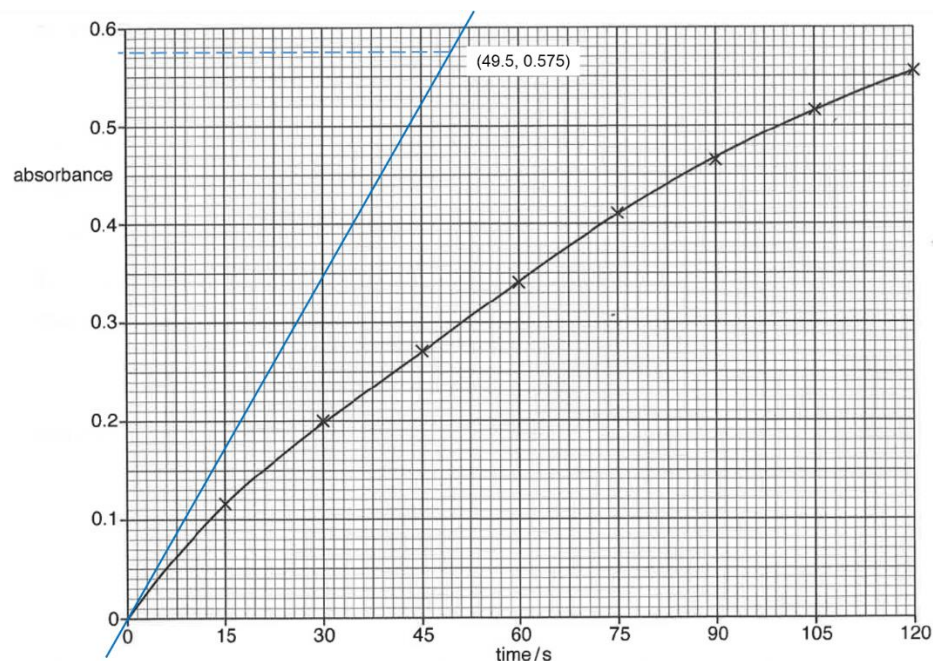
amount of Ag_2O required = 0.01667 mol

mass of Ag_2O required = $0.01667 \times (2 \times 107.9 + 16.0)$

$$= 3.864 \text{ g}$$

$$= 3.86 \text{ g (to 3 s.f.)}$$
 [1]

(c) (i)



$$\begin{aligned}\text{initial rate of reaction} &= \frac{0.575}{49.5} \\ &= 0.01162 \\ &= 0.0116 \text{ s}^{-1}\end{aligned}$$

[2]

(ii) The order of reaction is first order with respect to iodide ions. A linear graph is obtained when rate is plotted against $[I^-]$, showing that rate is directly proportional to $[I^-]$.

[1]

(iii) rate = $k[H_2O_2][I^-]$

[1]

(iv) rate = $k[H_2O_2][I^-]$

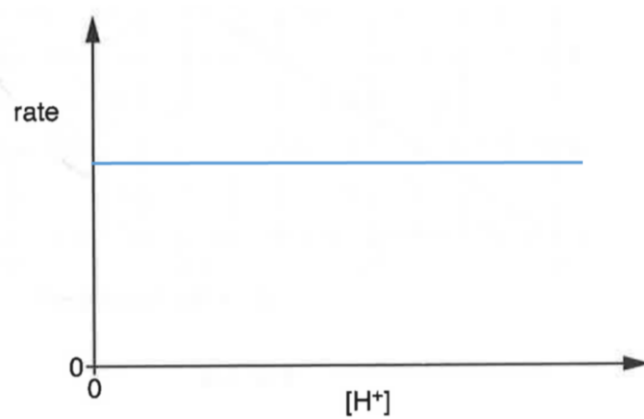
Since graph of rate against $[I^-]$ is a straight line,

gradient of line = $k[H_2O_2]$

$$5.30 = k \left(\frac{1.0 \times 0.1}{1.0 + 3.0 + 5.0 + 1.0} \right)$$

$$k = 530 \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

[2]

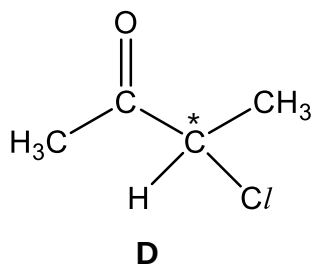
(v)**[1]****[Total: 12]**

- 3 (a) (i) The lone pair of electrons in the p orbital of oxygen atom is delocalised into the π electron cloud of the benzene ring, and thus less available to attack the carboxyl carbon. Hence, phenol is a weaker nucleophile than ethanol and does not react with carboxylic acids. [2]
- (ii) Step 1: NaOH(aq) (or aqueous sodium hydroxide)
Step 2: CH₃COCl (or ethanoyl chloride)
(Examiner's report: Many responses identified an incorrect reagent for step 2 or named the acid chloride incorrectly; some responses did not identify a specific reagent, e.g. "acyl chloride".) [2]
- (b) (i) Ethanolic KCN, heat under reflux [1]
- (ii) The carbocation formed in the S_N1 mechanism would be unstable as there is no electron-donating alkyl group attached to the electron deficient carbon to reduce the positive charge on the carbocation.
There is also an electron-withdrawing –COO[–] group that intensifies the positive charge on the electron deficient carbon.
(Examiner's report: Many responses gave answers in terms of the effect of the alkyl group on the stability of the carbocations. Weak responses referred to a carbocation with one alkyl group rather than none.) [3]
- (iii) The C atom in the C=O bond is less electron deficient due to the delocalisation of the negative charge over the –COO[–] group. Hence, the substitution occurs at the more electron deficient C atom bonded to an electronegative Cl atom. [1]

[Total: 9]

4 (a) (i) $\text{RCOCl} + \text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{HCl}$ [1]

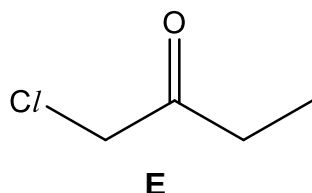
(ii)



(Examiners' report: the most common incorrect response did not account for the chiral centre on the molecule.)

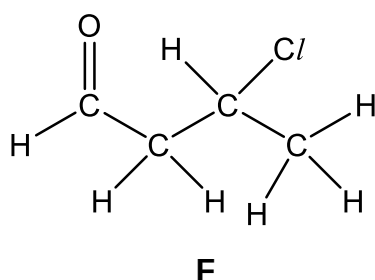
[2]

(iii)



[1]

(iv)



[2]

(b) (i) **G** and **H** are enantiomers of each other as they contain a chiral carbon and no internal plane of symmetry. [1]

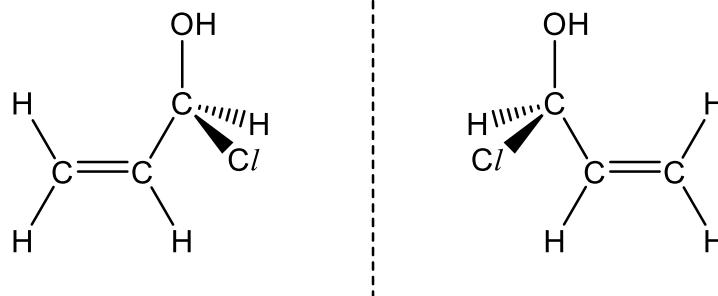
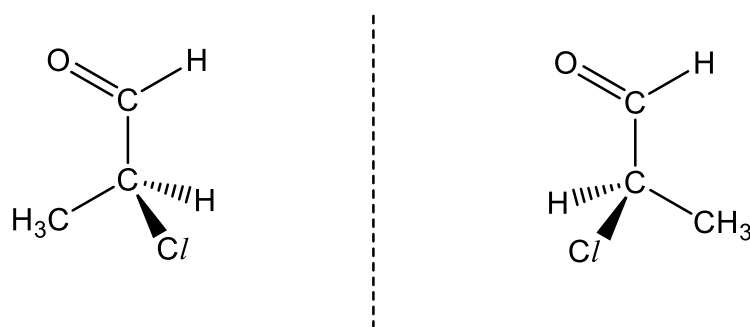
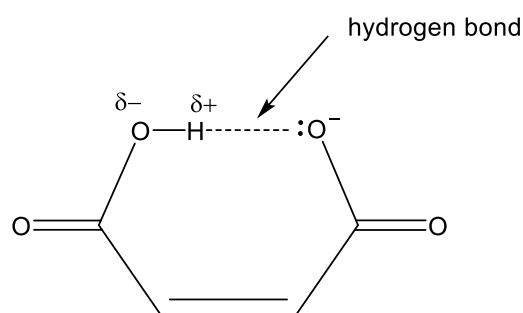
A mixture containing equal quantities of them is a racemic mixture. Since **G** and **H** rotate plane polarised light by the same extent in opposite directions, their effects on plane polarised light is cancelled out. [1]

A pure sample of G rotates plane polarised light in one direction, and this does not get cancelled off due to the absence of its enantiomer. [1]

(Examiners' report: weaker responses appreciated that enantiomers rotate the plane of polarisation of light in opposite directions, but not that the two effects cancelled each other.)

[3]

(ii)

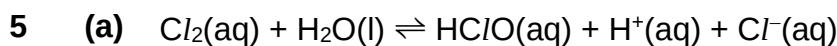
**G and H****or****[2]****(c) (i)**monoanion of *cis*-butenedioic acid**[1]**

(ii) There is restricted rotation about the C=C bond, and the –COOH group containing the protonic hydrogen is on the opposite side of the C=C bond from the lone pair of electrons on the O atom of the –COO[–] group. They are not in close proximity and direction to form intramolecular hydrogen bond in the monoanion of *trans*-butenedioic acid.

[2]

(iii) The monoanion of *cis*-butenedioic acid is more stable than that of *trans*-butenedioic acid due to intramolecular hydrogen bond. Hence, the position of equilibrium of $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ lies to the right, and *cis*-butenedioic acid is a stronger acid with a larger $K_{\text{a},1}$ value.

[1]**[Total: 15]**



$$K_c = \frac{[\text{HClO}][\text{H}^+][\text{Cl}^-]}{[\text{Cl}_2]}$$

$$\frac{[\text{HClO}]}{[\text{Cl}_2]} = \frac{K_c}{[\text{H}^+][\text{Cl}^-]}$$

$$= \frac{4.5 \times 10^{-4}}{10^{-7.0}(1.0 \times 10^{-3})}$$

$$= \frac{4.5 \times 10^6}{1}$$

[2]

(b) Similarity: Both HCl and HClO are monobasic acids.

Difference: HCl is a strong acid that dissociates completely in water while HClO is a weak acid that dissociates partially in water.

[2]

(c) mass of H_2O in $1 \text{ dm}^3 = 1.0 \times 1000$

$$= 1000 \text{ g}$$

$$\text{amount of } \text{H}_2\text{O} \text{ in } 1 \text{ dm}^3 = 1000 \div (2 \times 1.0 + 16.0)$$

$$= 55.56 \text{ mol}$$

$$\text{number of } \text{H}_2\text{O} \text{ molecules in } 1 \text{ dm}^3 = 55.56 \times 6.02 \times 10^{23}$$

$$= 3.344 \times 10^{25}$$

$$\text{number of free chlorine particles in } 1 \text{ dm}^3 \text{ at } 1 \text{ ppm} = (3.344 \times 10^{25}) \div 10^6$$

$$= 3.344 \times 10^{19}$$

$$\text{amount of free chlorine particles in } 1 \text{ dm}^3 \text{ at } 1 \text{ ppm} = \frac{3.344 \times 10^{19}}{6.02 \times 10^{23}}$$

$$= 5.556 \times 10^{-5} \text{ mol}$$

$$\text{concentration of free chlorine} = (5.556 \times 10^{-5}) \div 1$$

$$= 5.556 \times 10^{-5}$$

$$= 5.56 \times 10^{-5} \text{ mol dm}^{-3}$$

OR

$$\text{volume occupied by } 10^6 \text{ molecules of water} = \frac{10^6}{6.02 \times 10^{23}} \times \frac{(2 \times 1.0 + 16.0)}{1.0 \times 10^3}$$

$$= 2.990 \times 10^{-20} \text{ dm}^3$$

$$\text{concentration of free chlorine} = \frac{1}{6.02 \times 10^{23}} \div (2.990 \times 10^{-20})$$

$$= 5.56 \times 10^{-5} \text{ mol dm}^{-3}$$

[2]

(d) amount of $\text{S}_2\text{O}_3^{2-} = 4.00 \times 10^{-4} \times \frac{37.50}{1000}$
 $= 1.50 \times 10^{-5} \text{ mol}$

amount of $\text{I}_2 = 1.50 \times 10^{-5} \times \frac{1}{2}$
 $= 7.50 \times 10^{-6} \text{ mol}$

concentration of free chlorine $= 7.50 \times 10^{-6} \div \frac{150}{1000}$
 $= 5.00 \times 10^{-5} \text{ mol dm}^{-3}$

The quality of the sample of pool water is not acceptable as the concentration of free chlorine is lower than $5.56 \times 10^{-5} \text{ mol dm}^{-3}$ i.e. 1 ppm. [2]

(e) (i) $K_a = \frac{[\text{H}^+][\text{ClO}^-]}{[\text{HClO}]}$

Let $[\text{ClO}^-]$ be $x \text{ mol dm}^{-3}$.

$$3.7 \times 10^{-8} = \frac{(10^{-8.0})(x)}{6.0 \times 10^{-5} - x}$$

$$x = 4.723 \times 10^{-5}$$

$$[\text{ClO}^-] = 4.72 \times 10^{-5} \text{ mol dm}^{-3}$$

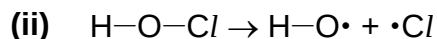
$$[\text{HClO}] = 6.0 \times 10^{-5} - 4.723 \times 10^{-5}$$

$$= 1.28 \times 10^{-5} \text{ mol dm}^{-3}$$

[3]

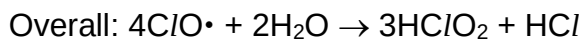
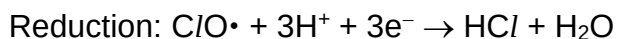
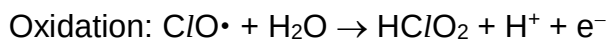
(ii) At a higher pH of 8.0, the position of equilibrium of the reaction shown in equation 5 will lie more to the right due to the lower $[\text{H}^+]$. Thus, $[\text{ClO}^-]$ will be higher and $[\text{HClO}]$ will be lower, resulting in a lower effectiveness of the free chlorine as a disinfectant. [2]

(f) (i) A free radical is a chemical species that contains at least one unpaired electron. [1]

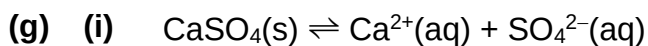


The $\text{O}-\text{Cl}$ bond in $\text{H}-\text{O}-\text{Cl}$ is more likely to break as less energy (203 kJ mol^{-1}) is required to overcome the weaker $\text{O}-\text{Cl}$ bond, compared to the 460 kJ mol^{-1} required to overcome the stronger $\text{H}-\text{O}$ bond. [2]



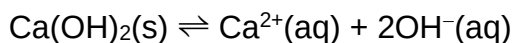


- (v) As the free chlorine decomposes in the presence of ultraviolet light, the position of equilibrium of the reaction between dichloroisocyanuric acid and OH^- will shift to the right to replenish the ClO^- lost and maintain the level of free chlorine. [1]



The addition of CaCl_2 increases $[\text{Ca}^{2+}(\text{aq})]$ and shifts the position of equilibrium to the left. Thus, the solubility of CaSO_4 decreases and prevents the plaster from dissolving. [2]

(ii) $[\text{OH}^-] \text{ at pH } 8.0 = 10^{14 - 8.0}$
 $= 1.00 \times 10^{-6} \text{ mol dm}^{-3}$



$K_{\text{sp}} = [\text{Ca}^{2+}][\text{OH}^-]^2$

$[\text{Ca}^{2+}] = \frac{5.5 \times 10^{-6}}{(1.00 \times 10^{-6})^2}$

$= 5.50 \times 10^6 \text{ mol dm}^{-3}$

The cloudiness is not likely to be due to the formation of solid calcium hydroxide as a very high $[\text{Ca}^{2+}]$ is required at pH 8.0 for precipitation of $\text{Ca}(\text{OH})_2$ to occur. [3]

[Total: 25]

Paper 3 Section A

1 (a) BF_3 : 120°

Explanation: 3 bonding pairs, 0 lone pair. Bonding pairs will minimise repulsion by adopting 120° angle to maximise distance from each other.

NH_3 : 107°

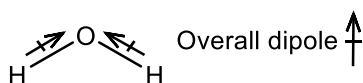
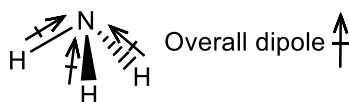
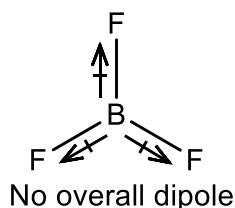
Explanation: 3 bonding pairs, 1 lone pair. Lone pair-bond pair repulsion is greater than bond pair-bond pair repulsion, which leads to a bond angle less than 109.5° .

H_2O : 104.5°

Explanation: 2 bonding pairs, 2 lone pairs. Lone pair-lone pair repulsion is greater than lone pair-bond pair repulsion, which leads to an even smaller bond angle (than 107°).

[4]

(b)



[4]

(c) (i) $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$

base acid

[1]

(ii) $:\text{NH}_3 + \text{BF}_3 \rightarrow \text{H}_3\text{N} \rightarrow \text{BF}_3$

base acid

[1]

(d) (i) Negligible forces of attraction between gas particles, and size of gas particles is negligible compared to the volume occupied by gas.

("High temperature and pressure" are not intrinsic properties of a gas but rather the conditions of the surrounding)

[1]

(ii) Stronger and more extensive hydrogen bonds exist between $\text{H}_2\text{O}(\text{g})$ molecules compared to $\text{NH}_3(\text{g})$ molecules, while the hydrogen bonds

[2]

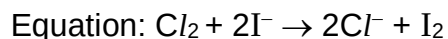
between $\text{NH}_3(\text{g})$ molecules are stronger than the instantaneous dipole-induced dipole interactions between $\text{BF}_3(\text{g})$ molecules.

- (e) (i) Volatility of Group 17 elements decrease down the group.

Down the group, there is an increasing number of electrons in each halogen molecule. This results in an increasing strength of instantaneous dipole-induced dipole interactions between halogen molecules, which requires more energy to overcome.

[2]

- (ii) $\text{Cl}_2(\text{aq})$ and $\text{KI}(\text{aq})$: Brown



$\text{Br}_2(\text{aq})$ and $\text{KCl}(\text{aq})$: Orange (no reaction)

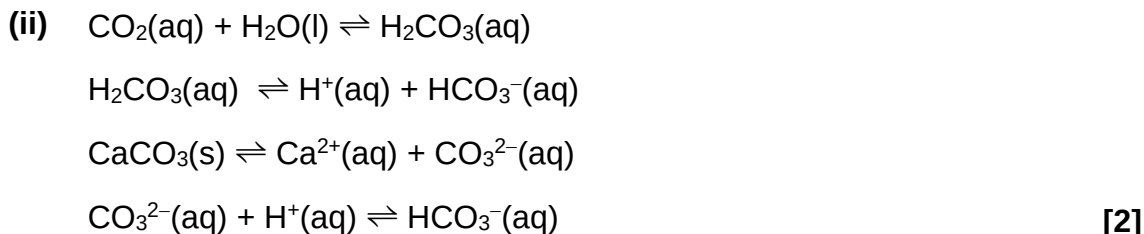
$\text{I}_2(\text{aq})$ and $\text{KBr}(\text{aq})$: Brown (no reaction)

(The question requires the predictions of final colours of the solutions, even if there were no reaction taking place)

[2]

[Total: 17]

2 (a) (i)
$$\frac{[\text{CO}_2]_{\text{sat. at 300 bar}}}{[\text{CO}_2]_{\text{sat. at 1 bar}}} = \frac{300}{1}$$
$$[\text{CO}_2]_{\text{sat. at 300 bar}} = 300 \times 0.040$$
$$= 12.0 \text{ mol dm}^{-3}$$
 [1]



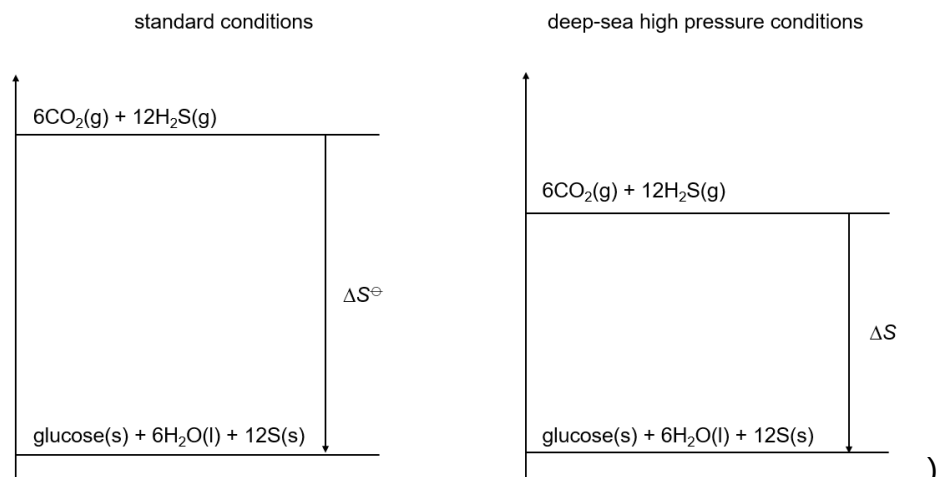
(b) $\Delta H^\ominus = [(-1273.3) + 6(-285.8) + 0] - [6(-393.5) + 12(-20.6)]$
 $= -379.9 \text{ kJ mol}^{-1}$
 $= -380 \text{ kJ mol}^{-1}$ (3 s.f.) [2]

(c) (i) $\Delta G^\ominus = \Delta H_r^\ominus - T\Delta S^\ominus$
 $-83 = -379.9 - 298\Delta S^\ominus$
 $\Delta S^\ominus = -0.996 \text{ kJ K}^{-1} \text{ mol}^{-1}$ [1]

(ii) ~~$[\text{CO}_2]$ is higher at deep-sea hydrothermal vents, thus position of equilibrium of $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq})$ lies more to the right, i.e. larger K_c . Since $\Delta G = -RT \ln K_c$ and K_c is larger, ΔG is more negative than -83 kJ mol^{-1} .~~

Under the high pressure conditions of deep-sea hydrothermal vents, the gas molecules are closer together with less ways to rearrange the gaseous molecules, the entropy of the gaseous particles decreases, resulting in a less negative ΔS . $-T\Delta S$ becomes less positive (more negative). Since $\Delta G = \Delta H - T\Delta S$, ΔG will become more negative.

(To help visualise:



[1]

Examiner's Comment: Many suggested that because $T\Delta S$ was negative, the value of ΔG would become more negative as T increased, forgetting that in the ΔG equation the term appears as $-T\Delta S$, and so the negative value of ΔS would cause ΔG to be more positive as T increases.

- (d) (i) A_r of sulfur sample = $\frac{(32 \times 93.5) + (33 \times 1.5) + (34 \times 4.5) + (36 \times 0.5)}{100}$
 $= 32.13$ (4 s.f.) [2]
- (ii) Half-life is the time taken for the concentration of the reactant, ^{35}S , to be halved. [1]
- (iii) It takes 3 half-lives for radioactivity to fall to 1/8 of its initial value.
time taken = 87×3
 $= 261$ days [1]
- (iv) $k = \frac{0.693}{87}$
 $= 0.007966 \text{ day}^{-1}$
 $\frac{(^{35}\text{S})_3}{(^{35}\text{S})_0} = 10^{\frac{0.007966 \times 3}{2.3}} = 0.9764$
Percentage decrease = $(1 - 0.9764) \times 100\%$
 $= 2.36\%$ [2]
- (e) (i)

Eqm conc / mol dm ⁻³	$\text{HS}^-(\text{aq})$	+	$\text{H}_2\text{O}(\text{l})$	$\rightleftharpoons$	$\text{H}_2\text{S}(\text{aq})$	+	$\text{OH}^-(\text{aq})$
	$0.1 - x$				x		x

$\text{p}K_b = 14 - 7.05$
 $= 6.95$
 $K_b = 10^{-6.95} \approx \frac{x^2}{0.10}$ (assuming x is very small)
 $[\text{OH}^-] = x = 0.0001059 \text{ mol dm}^{-3}$
 $\text{pOH} = -\lg(0.0001059)$
 $= 3.975$
 $\text{pH} = 14 - 3.975$
 $= 10.0$ [3]

(ii) A larger $\text{p}K_a$ means the acid is weaker with a lower extent of acid dissociation. [2]

$\text{CH}_3\text{CH}_2\text{SH}$ has the largest $\text{p}K_{\text{a}}$ because it is the weakest acid. The electron donating ethyl group intensifies the negative charge on S of $\text{CH}_3\text{CH}_2\text{S}^-$, destabilises the anion to a large extent, thus reduces the extent of acid dissociation.

$\text{C}_6\text{H}_5\text{SH}$ has the lowest $\text{p}K_{\text{a}}$ because it is the strongest acid. The negative charge on S of $\text{C}_6\text{H}_5\text{S}^-$ can be delocalised into the benzene ring, stabilises the anion to a large extent, thus increases the extent of acid dissociation.

- (iii) $BE(\text{S}-\text{H}) = 347 \text{ kJ mol}^{-1}$ $BE(\text{O}-\text{H}) = 410 \text{ kJ mol}^{-1}$

$\text{CH}_3\text{CH}_2\text{SH}$ is a stronger acid because less energy is required to break the weaker S—H bond in $\text{CH}_3\text{CH}_2\text{SH}$ for acid dissociation compared to the O—H bond in $\text{CH}_3\text{CH}_2\text{OH}$.

Or

S is less electronegative than O and does not draw as much electron clouds towards itself. As the negative charge on S of $\text{CH}_3\text{CH}_2\text{S}^-$ is not as intensified compared to the negative charge on O of $\text{CH}_3\text{CH}_2\text{O}^-$, $\text{CH}_3\text{CH}_2\text{S}^-$ is more stable and thus dissociation to a larger extent.

[1]

- (f) (i) More energy is required to overcome the stronger hydrogen bonds between $\text{CH}_3\text{CH}_2\text{OH}$ molecules compared to the weaker permanent dipole-permanent dipole (pd-pd) interactions between CH_3OCH_3 molecules. Thus boiling point of $\text{CH}_3\text{CH}_2\text{OH}$ is higher.

CH_3SCH_3 molecule has a larger number of electrons than CH_3OCH_3 molecule. More energy is required to overcome the stronger instantaneous dipole-induced dipole interactions between CH_3SCH_3 molecules compared to the weaker pd-pd interactions between CH_3OCH_3 molecules.

[2]

- (ii) Both compounds are relatively non-polar and have the same number of electrons in their molecules.

[1]

[Total: 22]



(ii) $E^\ominus_{\text{cell}} = E^\ominus(\text{ClO}^-/\text{Cl}^-) - E^\ominus(\text{IO}_3^-/\text{I}^-)$
 $= (+0.81) - (+0.26)$
 $= +0.55 \text{ V}$ [1]

(iii) $\Delta G^\ominus = -nFE^\ominus_{\text{cell}}$
 $= -(6)(96500)(+0.55)$
 $= -318450 \text{ J mol}^{-1}$
 $= -318 \text{ kJ mol}^{-1}$ [1]



$$\text{amount of } \text{S}_2\text{O}_3^{2-} = \frac{36.20}{1000} \times 0.100$$

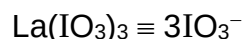
$$= 3.62 \times 10^{-3} \text{ mol}$$

$$\text{amount of } \text{I}_2 \text{ formed} = \frac{1}{2} \times 3.62 \times 10^{-3}$$

$$= 1.81 \times 10^{-3} \text{ mol}$$

$$\text{amount of } \text{La}(\text{IO}_3)_3 \text{ used} = \frac{1.00}{138.9 + (3 \times 126.9) + (9 \times 16.0)}$$

$$= 1.507 \times 10^{-3} \text{ mol}$$



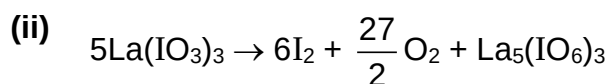
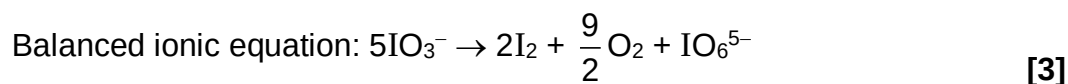
$$\text{amount of } \text{IO}_3^- = 3 \times 1.507 \times 10^{-3}$$

$$= 4.521 \times 10^{-3} \text{ mol}$$

Molar ratio of $\text{IO}_3^- : \text{I}_2$

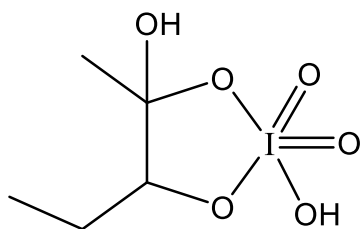
$$= 4.521 \times 10^{-3} : 1.81 \times 10^{-3}$$

$\approx 5 : 2$ (shown)



(Examiners' report: The majority of candidates did not work out the formula of lanthanum periodate(VII).) [2]

(c) (i)

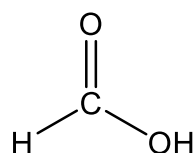
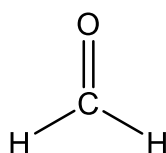
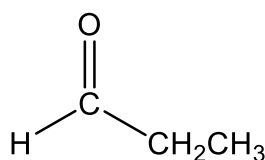


cyclic intermediate

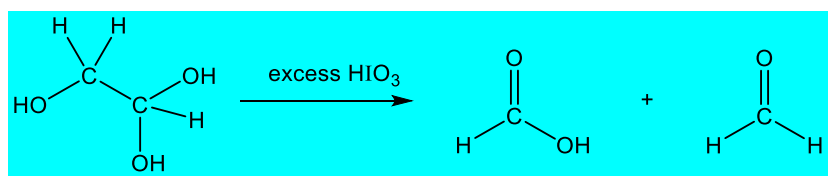
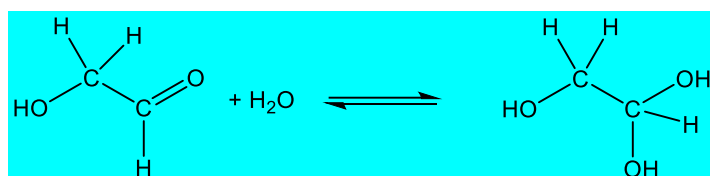
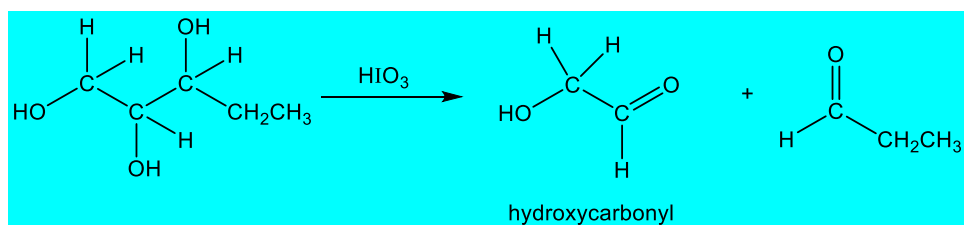
(Examiners' report: Most did not consider the conversion of the carbonyl group via the hydration equilibrium, but rather gave the intermediate that would have been obtained from the diol, 2,3-dihydroxypentane.)

[1]

(ii) Products from first reaction:

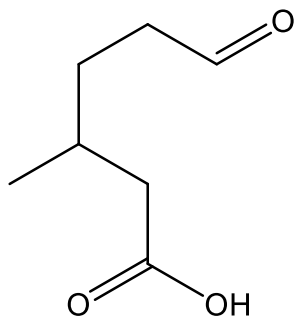


(Note:

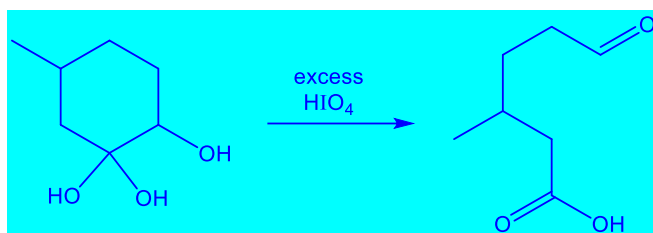
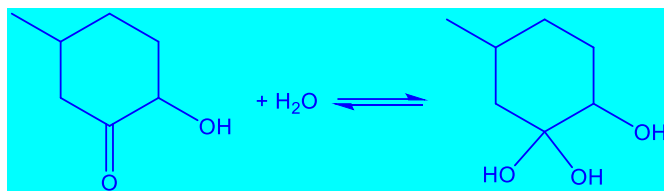


[3]

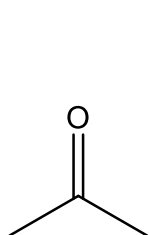
Product from second reaction:



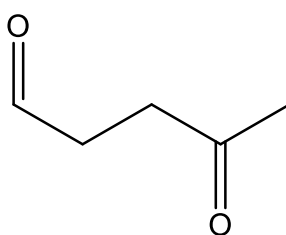
Note:



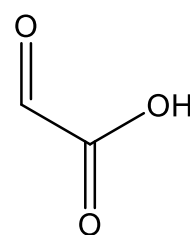
- (d) (i)
- Citral undergoes oxidation with Tollens' reagent
 $\Rightarrow$ citral is an aldehyde
 - Citral undergoes electrophilic addition with $\text{Br}_2(\text{aq})$
 $\Rightarrow$ citral is an alkene
 - Citral undergoes mild oxidation with cold KMnO_4 to form **A**
 $\Rightarrow$ **A** is a diol
 - **A** does not undergo oxidation with Tollens' reagent but undergoes acid-carbonate reaction with $\text{Na}_2\text{CO}_3(\text{aq})$
 $\Rightarrow$ **A** is not an aldehyde, **A** is a carboxylic acid
 - **A** undergoes selective oxidation with HIO_4 to give **B**, **C** and **D**
 $\Rightarrow$ **A** is an alcohol with at least 2 adjacent $-\text{OH}$ groups
 - **B** and **C** give positive iodoform test / undergo oxidation with alkaline aqueous iodine
 $\Rightarrow$ **B** and **C** contain $-\text{COCH}_3$ group or $-\text{CH}(\text{CH}_3)\text{OH}$ group
 - **C** and **D** undergo oxidation with Fehling's solution
 $\Rightarrow$ **C** and **D** are aliphatic aldehydes



B

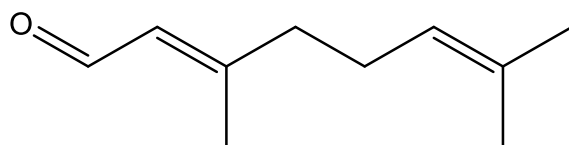


C



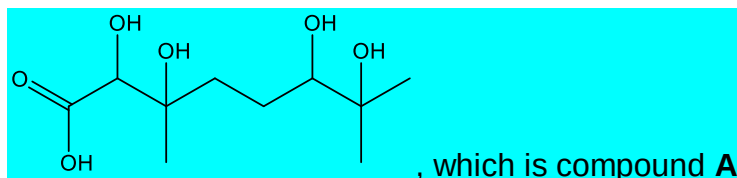
D

[7]

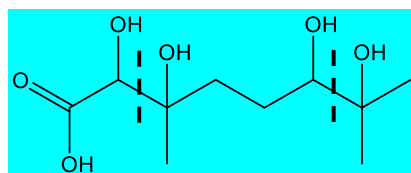


citral

Note: citral undergoes mild oxidation with cold KMnO_4 to form



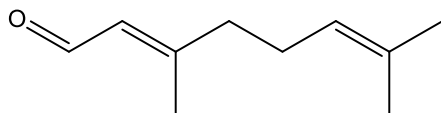
, which is compound A.



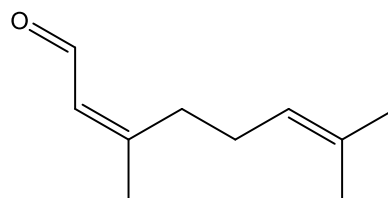
gives D gives C gives B

Note: Alternative structures can be deduced for citral, but they do not fulfill the criterion in (d)(ii).

- (ii) Cis-trans isomerism is present as there is restricted rotation about $\text{C}=\text{C}$ bond, and each C atom on the $\text{C}=\text{C}$ bond is bonded to different groups.



trans-isomer



cis-isomer

[2]

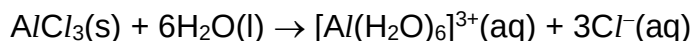
[Total: 21]

Paper 3 Section B

- 4 (a) NaCl has a giant ionic lattice structure and strong electrostatic attraction between Na^+ and Cl^- ions. NaCl dissolves in water to form a neutral solution of pH 7.



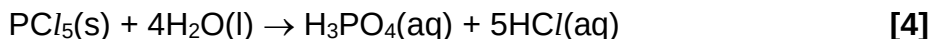
AlCl_3 has a simple covalent structure and dissolves in water.



Due to the small size and high charge, Al^{3+} is highly polarising. Al^{3+} is able to polarise the electron cloud of H_2O and weaken the O—H bonds in water, resulting in hydrolysis to give an acidic solution of pH 3.



PCl_5 has a simple covalent structure. PCl_5 dissolves and undergoes hydrolysis in water to form an acidic solution of pH 2. P atom in PCl_5 has energetically accessible vacant 3d orbitals to form dative bonds with water molecules.



(b) (i) $K_{\text{sp}} = [\text{Pb}^{2+}][\text{Cl}^-]^2$

units: $\text{mol}^3 \text{dm}^{-9}$

[1]



x

x

2x

$$K_{\text{sp}} = [\text{Pb}^{2+}][\text{Cl}^-]^2$$

$$1.7 \times 10^{-5} = 4x^3$$

$$x = 0.016198$$

$$[\text{Pb}^{2+}] = 0.016198 \text{ mol dm}^{-3}$$

$$\text{amount of PbCl}_2 = \text{amount of Pb}^{2+}$$

$$= 0.016198 \times \frac{10}{1000}$$

$$= 1.6198 \times 10^{-4} \text{ mol}$$

$$\text{mass of PbCl}_2 = 1.6198 \times 10^{-4} \times (207.2 + 2 \times 35.5)$$

$$= 0.0451 \text{ g}$$

$$K_{\text{sp}} = [\text{Pb}^{2+}][\text{I}^-]^2$$

$$9.8 \times 10^{-9} = 4x^3$$

$$x = 1.348 \times 10^{-3}$$

[3]

$$[\text{Pb}^{2+}] = 1.348 \times 10^{-3} \text{ mol dm}^{-3}$$

amount of PbI_2 = amount of Pb^{2+}

$$= 1.348 \times 10^{-3} \times \frac{10}{1000}$$

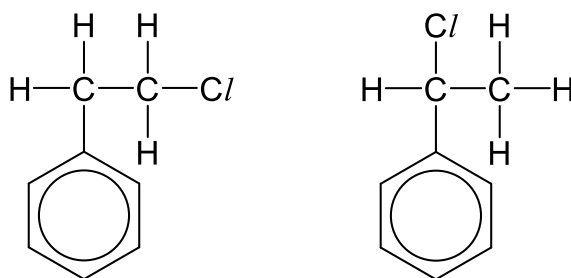
$$= 1.348 \times 10^{-5} \text{ mol}$$

$$\text{mass of } \text{PbI}_2 = 1.348 \times 10^{-5} \times (207.2 + 2 \times 126.9)$$

$$= 6.21 \times 10^{-3} \text{ g}$$

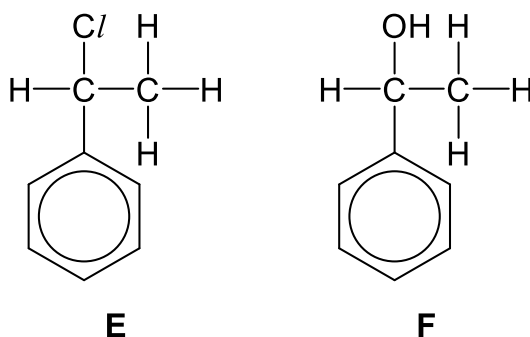
Hence, saturated solution of PbCl_2 will produce a larger mass of solid.

(c) (i)



[1]

(ii)



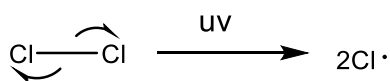
[1]

(iii) The formation of **E** proceeds via electrophilic addition mechanism and the formation of **F** proceeds via $\text{S}_{\text{N}}1$ mechanism. Both mechanisms involves the formation of trigonal planar carbocation intermediate. The nucleophile can attack the planar carbocation with equal probability from either side of the plane. Since both **E** and **F** are chiral, racemic mixtures are obtained which consist of 50:50 mixture of enantiomers.

[2]

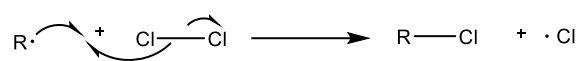
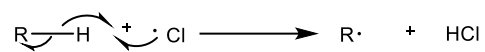
(d) (i) Mechanism: Free radical substitution

Initiation

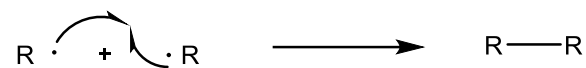
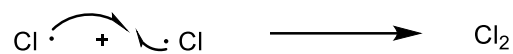


Propagation

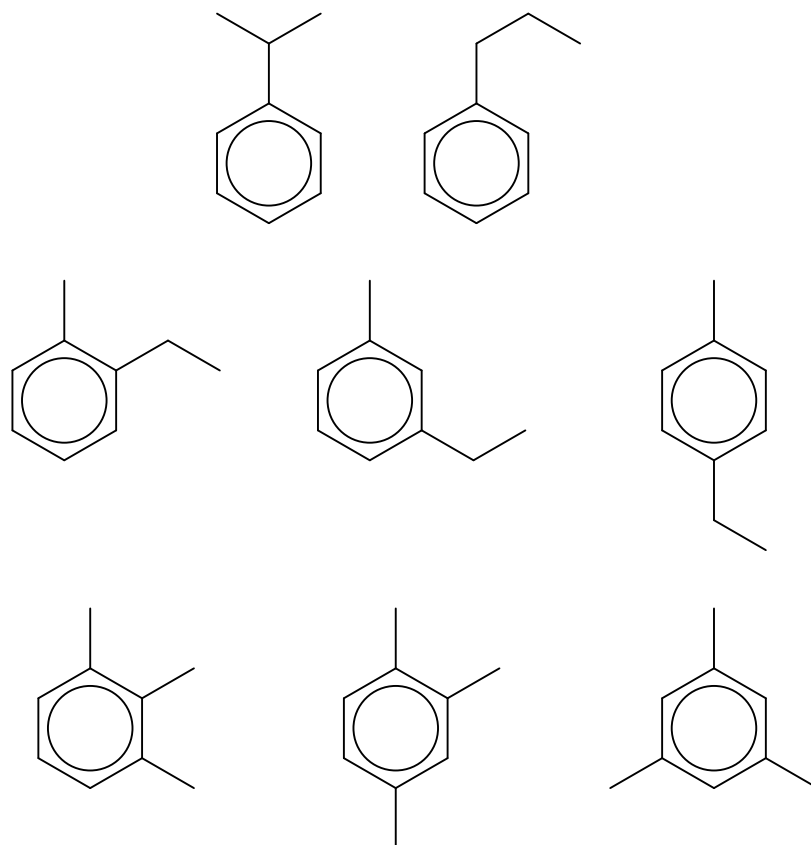
[3]



Termination

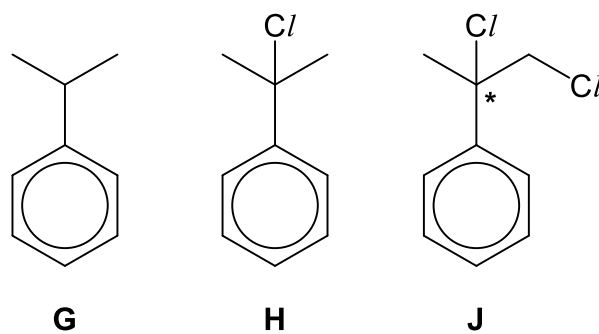


(ii)



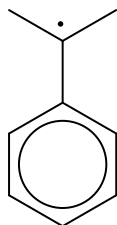
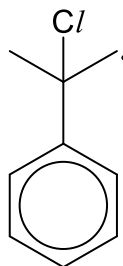
[2]

(iii)



[2]

(iv)

**Intermediate of G****Intermediate of H**

The radical intermediate formed in the reaction of **G** to form **H** is more stable as the single unpaired electron can be delocalised into the benzene ring. Thus, rate of reaction with Cl_2 is faster from **G** to **H**, then from **H** to **J**.

[1]**[Total: 20]**

5 (a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ [1]

(b) (i) The word degenerate means having the same energy. [1]

- (ii) In an isolated metal ion in the gaseous phase, the five d orbitals are degenerate. In an octahedral complex, the central metal ion is surrounded by six lone pairs of electrons (on the ligands). As a result, all five d orbitals experience electrostatic repulsion as the ligands approach and increase in energy. The degeneracy of the five d orbitals is also lost, and they are split into two energy levels, with an energy gap between them. The $d_{x^2-y^2}$ and d_{z^2} orbitals are at a higher energy level while the d_{xy} , d_{xz} and d_{yz} orbitals are at a lower energy level. This is because the $d_{x^2-y^2}$ and d_{z^2} orbitals have lobes that point towards the ligands, which results in greater repulsion compared to the other d orbitals.

(Examiner's report: Several candidates had learned this area of the syllabus solely in the context of colour of complexes and thus gave a variety of irrelevant statements about electrons being promoted from one orbital to another by absorbing a photon.)

[3]

(c) (i) Compound L: $\text{BrCH}_2\text{CH}_2\text{NHCH}_2\text{CH}_2\text{Br}$

Compound M: $\text{NCCH}_2\text{CH}_2\text{NHCH}_2\text{CH}_2\text{CN}$

Step 1: limited ethanolic NH_3 , heat in sealed tube

Step 2: ethanolic KCN, heat under reflux

Step 3: $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux

[4]

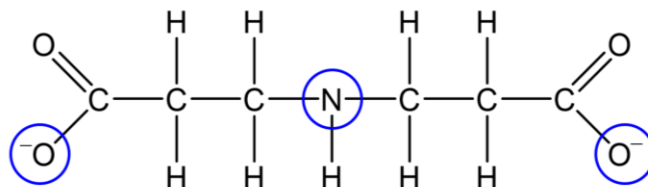
- (ii) A possible organic side product is $\text{N}(\text{CH}_2\text{CH}_2\text{Br})_3$. To minimise its formation, shorter reaction time should be used.

OR

A possible organic side product is $\text{BrCH}_2\text{CH}_2\text{NH}_2$. To minimise its formation, excess dibromoethane should be used.

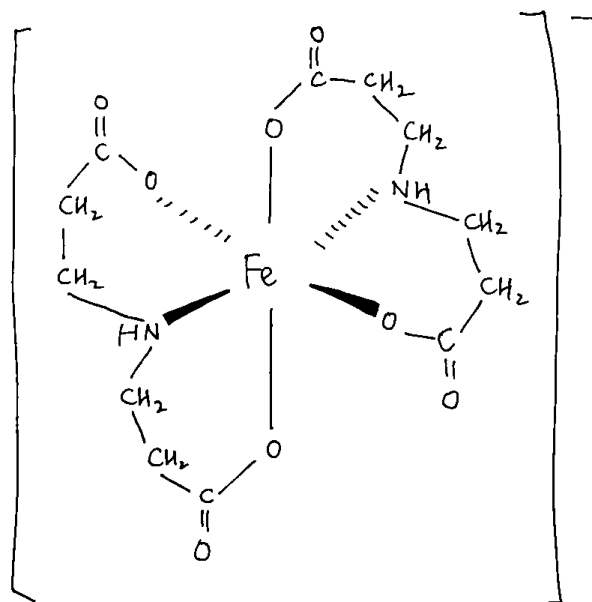
[2]

(iii)



[1]

(iv)



(Examiner's report: Most appreciated that the overall charge would be 1-; only a few of the attempts at drawing the 3-dimensional structure were correct.)

[2]

- (v) As CN^- is a stronger ligand than H_2O , it splits the d orbitals by a greater extent. Thus, the d electrons will fill up the d orbitals in the lower energy level before filling up the d orbitals in the higher energy level. This is because the energy for promotion of d electron from lower to higher energy level is lower than the energy required to overcome electronic repulsion.

[1]

(d) (i)

$$K_p = \frac{(p_{\text{CO}_2})^3}{(p_{\text{CO}})^3}$$

no units

(Examiner's report: Quite a number did not state explicitly that K_p is dimensionless.)

[1]

(ii)

	$\text{Fe}_2\text{O}_3(\text{s})$	+	$3\text{CO}(\text{g})$	$\rightleftharpoons$	$2\text{Fe}(\text{l})$	+	$3\text{CO}_2(\text{g})$
Initial partial pressure / bar	—		1.00		—		0
Change in partial pressure / bar	—		$-3x$		—		$+3x$
Equilibrium partial pressure / bar	—		$1.00 - 3x$		—		$3x$

$$K_p = \frac{(3x)^3}{(1.00 - 3x)^3} = 19.9$$

[2]

$$\frac{3x}{1.00 - 3x} = 2.710$$

$$x = 0.2435$$

$$\begin{aligned}\text{Equilibrium partial pressure of CO}_2 &= 3 \times 0.2345 \\ &= 0.730 \text{ bar}\end{aligned}$$

(Examiner's report: Many candidates gave the answer as the cube root of 19.9, i.e. 2.71.)

- (iii) The entropy change for this reaction is small but positive as there is a change in phase from solid Fe_2O_3 to liquid Fe. The volume change is negligible but there is an increase in disorder as the orderly arrangement of particles in the solid is destroyed. There are more ways to arrange the particles and distribute the energy in liquid state than in solid state.

[2]

[Total: 20]